

Unit 5 – Sampling Distributions

Notes 5 – Sampling Distribution of a Difference in Sample Means

Why Do We Have This?

- Remember this is common practice in statistical inference!
- We look at two samples from two populations to see if there is a significant difference
- We look at two experimental treatments to see if the difference is due to natural variation or the treatment themselves
- These problems will all be looked at in the coming units!

Recall: Sampling Distribution of $\bar{x}$

Shape	Center	Spread
As long as...	As long as...	As long as...

Sampling Distribution of $\bar{x}_1 - \bar{x}_2$

_____ = sample mean from the first group

_____ = sample mean from the second group

Shape	Center	Spread

As long as...	As long as....	As long as...

Example) A study is conducted to compare the effectiveness of two study techniques, A and B, in improving test scores. For Technique A, a random sample of 80 students yields an average score of 85 with a standard deviation of 10. For Technique B, a random sample of 100 students yields an average score of 82 with a standard deviation of 8. Calculate the probability that the difference in the average scores between Technique A minus Technique B is greater than 4.